

AN OPEN SOURCE SOFTWARE PACKAGE TO COMPARE AND EXTEND EFFECTIVE MASS MODELS OF THE PHOSPHORUS DONOR IN SILICON

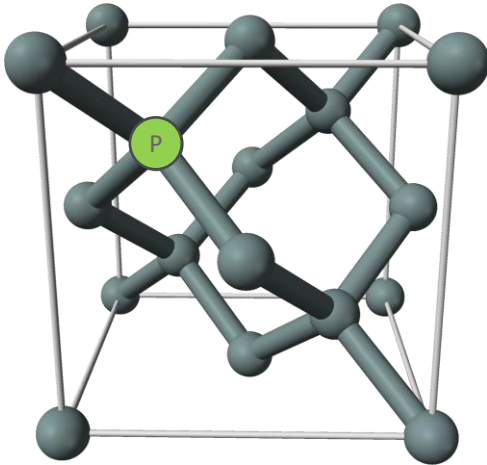
Luke Pendo and Xuedong Hu

Presentation Date: Monday, March 6, 2023 4:36 pm

gitlab.com/cantankerous-impurity

OUR PROBLEM

System description



- Infinite silicon crystal bulk.
- One silicon replaced by an isolated pnictogen (Series V/Group 15).
- Specifically we consider the **phosphorous** substitutional donor.
- System is symmetric about the impurity under **operations of point group T_d** .
- Excess electric charge of phosphorous nuclear attracts excess electron.

Original crystal structure picture by Ben Mills, public domain.

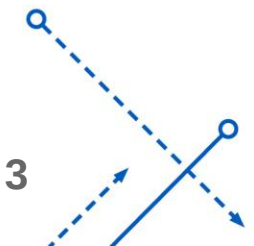
<https://commons.wikimedia.org/w/index.php?curid=2722584>

Modified by L. Pendo for presentation.

The back of the envelope idea

- Excess electron is affected by a **periodic mean field potential**, and **weakly bound to donor potential**.
 - **Short-range**: looks like a **Bloch wave**.
 - **Long-range**: **attenuated on a radial scale** much longer than lattice parameter.

[https://gitlab.com/cantankerous-impurity/Documentation/-/tree/core/Presentations/MarchMeeting2023\(LasVegas\)](https://gitlab.com/cantankerous-impurity/Documentation/-/tree/core/Presentations/MarchMeeting2023(LasVegas))



THE MODEL

The effective mass approximation

EMA: Assume the donor electron is localized in $\mathbf{k}$ -space at the **six-fold degenerate minima** of the silicon conduction band.

$$|\Psi\rangle = \sum_n \int_{BZ} d\mathbf{k} A_{\mathbf{k}} |n\mathbf{k}\rangle \rightarrow \sum_{\mathbf{k}_p} \beta_{\mathbf{k}_p} \int_{BZ} d\mathbf{k} A_{\mathbf{k}}^{(p)} |0\mathbf{k}\rangle$$

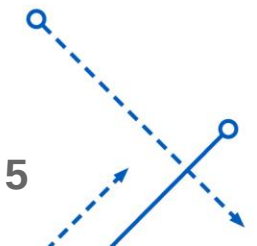
$$\Psi_{\beta;\mathbf{r}} \approx \sum_{\mathbf{k}_p} \beta_{\mathbf{k}_p} F_{\mathbf{r}}^{(p)} \Theta_{\mathbf{r}}^{(p)}$$

$$F_{\mathbf{r}}^{(p)} := \int_{BZ} d\mathbf{k} A_{\mathbf{k}}^{(p)} e^{i(\mathbf{k}-\mathbf{k}_p)\cdot\mathbf{r}} \quad \Theta_{\mathbf{r}}^{(p)} := e^{i\mathbf{k}_p\cdot\mathbf{r}} u_{\mathbf{k}_q\mathbf{r}}$$

- *Short-range:* Bloch wave.
- *Long-range:* Envelope function.

We expand the envelope function with a set of basis functions.

$$F_{\mathbf{r}}^{(q)} = \sum_j c_j \Phi_{j\mathbf{r}}^{(q)}$$



The Generalized Eigenvalue Problem

“Modified” Shindo-Nara effective mass equation[1] cast as a *generalized eigenvalue problem*[2].

$$E\mathcal{S}c = (\mathcal{T} + \mathcal{U})c$$

$$c^\dagger \mathcal{S}c := \langle \Psi_\beta | \Psi_\beta \rangle \quad c^\dagger \mathcal{T}c := \langle \Psi_\beta | \hat{H}_0 | \Psi_\beta \rangle \quad c^\dagger \mathcal{U}c := \langle \Psi_\beta | \hat{U} | \Psi_\beta \rangle$$

H_0 is intended to model the pure silicon potential.

U is intended to model *all* deviations from the pure silicon potential.

symmetrized anisotropic basis states

$$\Psi_{\beta\mathbf{r}} \approx \sum_j c_j \left(\sum_{q \in \{x,y,z\}} \beta_q \Phi_{j\mathbf{r}}^{(q)} \Theta_{\Omega\mathbf{r}}^{(q)} \right)$$

Bloch functions of *opposed valley minima* are included as Fourier series: $\Theta_{\Omega\mathbf{r}}^{(q)} := \frac{\Theta_{\mathbf{r}}^{(q)} + \Omega \Theta_{\mathbf{r}}^{(-q)}}{\sqrt{2}}$



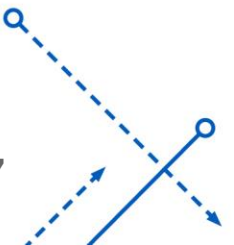
Matrix Construction – Overlap, Hamiltonian

$$\mathcal{S}_{ij} = \sum_{pq} \beta_p^* \beta_q \int d\mathbf{r} e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \tilde{\Phi}_{i\mathbf{r}}^{(p)} \Phi_{j\mathbf{r}}^{(q)}$$

$$\mathcal{T}_{ij} = \sum_{pq} \beta_p^* \beta_q \int d\mathbf{r} e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \tilde{\Phi}_{i\mathbf{r}}^{(p)} \left[\hat{T}^{(q)} \Phi_{j\mathbf{r}}^{(q)} \right]$$

$$\hat{T}^{(z)} = - \left(\frac{\hbar^2}{2m_{\perp}} \right) \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{m_{\perp}}{m_{\parallel}} \frac{\partial^2}{\partial z^2} \right)$$

$$\mathcal{U}_{ij} = \sum_{pq} \beta_p^* \beta_q \int d\mathbf{r} \tilde{\Theta}_{\Omega_i \mathbf{r}}^{(p)} \Theta_{\Omega_j \mathbf{r}}^{(q)} \tilde{\Phi}_{i\mathbf{r}}^{(p)} \Phi_{j\mathbf{r}}^{(q)} U_r$$



Views on the effective mass approximation

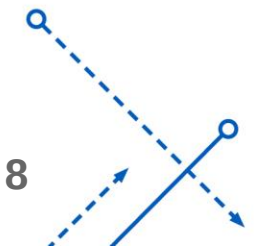


- An attractively intuitive physical picture of the state of the donor electron.
- Less computationally expensive as compared to *ab initio* and tight-binding models.
- Predicts the energies of spatially extended excited states of donors [1,2].



- Struggles to present a quantitatively persuasive picture of the ground state manifold
- Few agreed upon criteria for refutability (beyond agreement with energy spectrum).
- Lots of freedom to generate different schemes through reasonable but ad hoc assumptions.

Is further improvement of effective mass methods possible?



Model evaluation

Out of model

Verisimilitude

How well does the model resemble reality?

In-model

Consistency

How well does the model play by its own rules?

Completeness

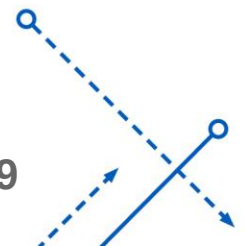
Can the model answer questions posed to it?

V-score[*]: Metric for model completeness

Is the trial eigenstate **well-converged** to an actual eigenstate?

$$v^2 := \frac{\langle \hat{H}^2 \rangle}{E^2} - 1$$

$$\langle \hat{H}^2 \rangle := \frac{\langle \Psi_\beta | \hat{H}_0^2 + \hat{H}_0 \hat{U} + \hat{U} \hat{H}_0 + \hat{U}^2 | \Psi_\beta \rangle}{\langle \Psi_\beta | \Psi_\beta \rangle}$$



“Modified” Shindo-Nara equation?

$$\begin{aligned}
 \langle \Psi_\beta | \hat{H}_0^2 | \Psi_\beta \rangle &\approx \sum_{pq} \beta_p^* \beta_q \int_{BZ} d\mathbf{k} d\mathbf{k}' E_{\mathbf{k}} E_{\mathbf{k}'} \tilde{A}_{\mathbf{k}}^{(p)} A_{\mathbf{k}'}^{(q)} \delta(\mathbf{k}' - \mathbf{k}) \\
 &= \sum_{pg} \beta_p^* \beta_q \int d\mathbf{r} e^{i(\mathbf{k}_p - \mathbf{k}_q) \cdot \mathbf{r}} \left[\hat{T}^{(p)} F^{(p)} \right]^\dagger \left[\hat{T}^{(q)} F^{(q)} \right]
 \end{aligned}$$

It is a reasonable assumption for **verisimilitude** that the kinetic term **doesn't mix the valleys**.

$$\langle \Psi_\beta | \hat{H}_0^2 | \Psi_\beta \rangle \rightarrow \int d\mathbf{r} \left[\hat{T}^{(z)} F^{(z)} \right]^\dagger \left[\hat{T}^{(z)} F^{(z)} \right] \quad (?)$$

However, to ensure that the square of the V-score is **positive definite**, the previously dropped terms **must be kept**.

$$\langle \Psi_\beta | \hat{H}_0^2 | \Psi_\beta \rangle \rightarrow \sum_{pg} \beta_p^* \beta_q \int d\mathbf{r} e^{i(\mathbf{k}_p - \mathbf{k}_q) \cdot \mathbf{r}} \left[\hat{T}^{(p)} F^{(p)} \right]^\dagger \left[\hat{T}^{(q)} F^{(q)} \right]$$

Matrix Construction – Hamiltonian Squared

$$c^\dagger (\mathcal{T}^2) c := \langle \Psi_\beta | \hat{H}_0^2 | \Psi_\beta \rangle \quad c^\dagger (\mathcal{U}^2) c := \langle \Psi_\beta | \hat{U}^2 | \Psi_\beta \rangle$$

$$c^\dagger (\mathcal{T}\mathcal{U}) c := \langle \Psi_\beta | \hat{H}_0 \hat{U} | \Psi_\beta \rangle \quad c^\dagger (\mathcal{U}\mathcal{T}) c := \langle \Psi_\beta | \hat{U} \hat{H}_0 | \Psi_\beta \rangle$$

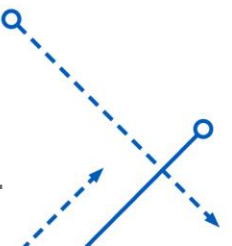
$$(\mathcal{T}^2)_{ij} = \sum_{pq} \beta_p^* \beta_q \int d\mathbf{r} e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \left[\hat{T}^{(p)} \Phi_{i\mathbf{r}}^{(p)} \right]^\dagger \left[\hat{T}^{(q)} \Phi_{j\mathbf{r}}^{(q)} \right]$$

$$(\mathcal{U}^2)_{ij} = \sum_{pq} \beta_p^* \beta_q \int d\mathbf{r} \tilde{\Theta}_{\Omega_i \mathbf{r}}^{(p)} \Theta_{\Omega_j \mathbf{r}}^{(q)} \tilde{\Phi}_{i\mathbf{r}}^{(p)} \Phi_{j\mathbf{r}}^{(q)} U_{\mathbf{r}}^2$$

$$(\mathcal{T}\mathcal{U})_{ij} = \sum_{pq} \beta_p^* \beta_q \int d\mathbf{r} \tilde{\Theta}_{\Omega_i \mathbf{r}}^{(p)} \Theta_{\Omega_j \mathbf{r}}^{(q)} \left[\hat{T}^{(p)} \Phi_{i\mathbf{r}}^{(p)} \right]^\dagger \Phi_{j\mathbf{r}}^{(q)} U_{\mathbf{r}}$$

$$(\mathcal{U}\mathcal{T})_{ij} = \sum_{pq} \beta_p^* \beta_q \int d\mathbf{r} \tilde{\Theta}_{\Omega_i \mathbf{r}}^{(p)} \Theta_{\Omega_j \mathbf{r}}^{(q)} \tilde{\Phi}_{i\mathbf{r}}^{(p)} \left[\hat{T}^{(q)} \Phi_{j\mathbf{r}}^{(q)} \right] U_{\mathbf{r}}$$

$$\langle \hat{H}^2 \rangle = \frac{c^\dagger (\mathcal{T}^2 + \mathcal{T}\mathcal{U} + \mathcal{U}\mathcal{T} + \mathcal{U}^2) c}{c^\dagger \mathcal{S} c}$$



THE BASIS

Naïve basis functions defined

$$\Phi_{\lambda\eta;m\zeta lk}^{(Wq)} = \hat{Q}_{m\zeta}^{(q)} \hat{\mathcal{P}}_{\lambda;lm}^{(q)} \mathcal{R}_{\eta q;kl}^{(W)}$$

$$\vartheta_{\lambda\{\chi\}} = \sqrt{1 + (\lambda^2 - 1) \chi^2}$$

$$\hat{\mathcal{P}}_{\lambda;lm}^{(q)} = \sqrt{\frac{\lambda}{\vartheta_{\lambda\{\chi_q\}}^3}} \hat{P}_{lm}\{\chi'_q\}$$

$$\hat{Q}_{m\zeta}^{(q)} = \frac{1}{\sqrt{2\pi}} \left(\frac{e^{im\phi_q} + \zeta e^{-im\phi_q}}{2\delta_{m0} + \sqrt{2}(1 - \delta_{m0})} \right)$$

$$\chi_x = \frac{x}{r} = \sin(\theta) \cos(\phi)$$

$$\chi_y = \frac{y}{r} = \sin(\theta) \sin(\phi)$$

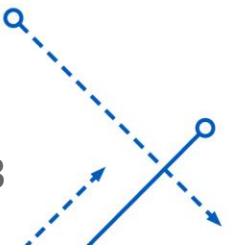
$$\chi_z = \frac{z}{r} = \cos(\theta)$$

$$\phi_{q_1} = \text{Arg}(\chi_{q_2} \pm i\chi_{q_3})$$

q_1	q_2	q_3
x	y	z
y	z	x
z	x	y

$$\eta_q = \frac{\eta}{\vartheta_{\lambda\{\chi_q\}}} \quad \chi'_q = \frac{\lambda\chi_q}{\vartheta_{\lambda\{\chi_q\}}}$$

$$\mathcal{R}_{\eta;kl}^{(W)} = \left(\frac{2}{W a_{\perp} \eta} \right)^{\frac{3}{2}} \left[\sqrt{W} \rho^{\frac{l}{W}} e^{-\rho/2} \hat{L}_{k\{\rho\}}^{\left(\frac{2l+3-W}{W}\right)} \right] \Big|_{\rho \rightarrow \left(\frac{2r}{W a_{\perp} \eta}\right)^W}$$



Complete, Extensible “Subbasis”

- Underlying naïve subbases are **orthonormal**, **fully complete** over $L^2(\mathbb{R}^3)$.
- *Solid angle* expansion in **spherical harmonics**. *Radial* expansion in **Laguerre polynomials**.
- Basis can utilize **Slater** (“hydrogen-like”) or **Gaussian** (“harmonic oscillator”) exponential decay.

$$\varphi_{m\zeta lk}^{(W)} = \sqrt{W} r^l \exp\left(-\frac{r^W}{2}\right) \hat{L}_{k\{r^W\}}^{\left(\frac{2l+3}{W}-1\right)} \left(\frac{\hat{Y}_l^{+m} + \zeta \hat{Y}_l^{-m}}{2\delta_{m0} + \sqrt{2}(1 - \delta_{m0})} \right)$$

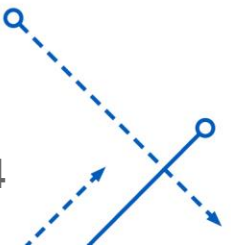
$$m = 0, 1, 2, \dots \quad \zeta = \pm 1 \quad l = m, m+1, m+2, \dots \quad k = 0, 1, 2, \dots$$

- *Variational* tuning: **anisotropy** along azimuth, overall **radial** scaling.
- **Bloch** functions of *valley minima* are included as Fourier series.

$$\Phi_{\lambda\eta; m\zeta lk, \Omega}^{(Wq)} = \sqrt{\lambda \left(\frac{2}{W\eta}\right)^3} \left(\varphi_{m\zeta lk}^{(W)} \middle| \begin{array}{l} r \rightarrow 2r/W\eta \\ z \rightarrow \lambda z \\ z \rightarrow q: \{x, y, z\} \end{array} \left(\frac{e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{\mathbf{k}_q} + \Omega e^{-i\mathbf{k}_q \cdot \mathbf{r}} u_{-\mathbf{k}_q}}{\sqrt{2}} \right) \right)$$

$$\Omega = \pm 1$$

- Users can freely select size, composition of each subbasis.



“Superbasis”

- Difficult to optimize an orthonormal, complete basis span over multiple energy eigenstates.
- This can be resolved by using a basis constructed as a concatenation of multiple subbases.
- *Superbasis* formed from multiple orthonormal *subbases* individually equipped with tuning parameters.
- *Superbasis* is ***non-orthonormal***, ***overcomplete*** on $L^2(\mathbb{R}^3)$.
- Optimization of each subbasis seems to greatly reduce numerical effects from overcomplete basis.
- **Total number and size of subbases limited only by computation limits.**
- **Work on-going to increase code efficiency to allow for larger sub- and super-bases.**

Complete, Extensible Basis

Representation of single eigenstate

- Underlying naïve subbases are *orthonormal*, *fully complete* over $L^2(\mathbb{R}^3)$.
- Two *variational* parameters: *anisotropy* along azimuth, overall *radial* scaling.
- **Users can select membership and size of each *subbasis*.**
- However, difficult to optimize *subbasis* to span over multiple energy eigenstates.

Representation of multiple eigenstates / energy spectra

- *Superbasis* concatenation of individual independent *subbases*.
- *Superbasis* *non-orthonormal*, *overcomplete* on $L^2(\mathbb{R}^3)$.
- Optimization of each *subbasis* seems to mitigate problems from overcompleteness.
- **Total number and composition of subbases limited ONLY by computation limits.**
- Work on-going to increase code efficiency to allow for larger sub- and super-bases.

Generalized basis states -- I

$$\Phi_{\lambda\eta;m\zeta lk}^{(Wq)(\kappa)} := \left[(1 - \kappa) \hat{I} + \kappa \hat{T}^{(q)} \right] \Phi_{\lambda\eta;m\zeta lk}^{(Wq)}$$

$$\Phi_{\lambda\eta;m\zeta lk}^{(Wq)(\kappa)} = \left(\frac{\hbar^2}{2m_0 a_{\perp}^2 \eta^2} \right)^{\kappa} \hat{Q}_{m\zeta}^{(q)} \sum_{s=-\kappa}^{+\kappa} \mathfrak{H}_{\lambda;lm,s}^{(\kappa)} \hat{\mathcal{P}}_{\lambda;(l+2s)m}^{(q)} \mathcal{R}_{\eta q;kl,s}^{(W)(\kappa)}$$

$$\mathcal{R}_{\eta;kl,s}^{(W)(\kappa)} = \left(\frac{2}{W a_{\perp} \eta} \right)^{\frac{3}{2}} \left[\sqrt{W} \rho^{\frac{l}{W} - \kappa \delta_{W1} - \delta_{s\bar{1}}} e^{-\rho/2} \sum_{v=-2\kappa}^{\kappa + \delta_{s\bar{1}}} \mathfrak{h}_{lk,sv}^{(\kappa)} \hat{L}_{(k+v)\{\rho\}}^{(\alpha + \kappa - \delta_{s\bar{1}})} \right]_{\rho \rightarrow \left(\frac{2r}{W a_{\perp} \eta} \right)^W}$$

Generalized basis states -- II

$$\mathfrak{H}_{\lambda;lm,s}^{(\kappa)} = \delta_{s0} - \kappa (1 - \gamma \lambda^2) \mathfrak{Y}_{lm,s}$$

$$\gamma = \frac{m_{\perp}}{m_{\parallel}}$$

$$\mathfrak{Y}_{lm,0} = \frac{2l^2 - 2m^2 + 2l - 1}{(2l - 1)(2l + 3)}$$

$$\mathfrak{Y}_{lm,-1} = \sqrt{\frac{((l - 1)^2 - m^2)(l^2 - m^2)}{(2l - 3)(2l - 1)^2(2l + 1)}}$$

$$\mathfrak{Y}_{lm,+1} = \sqrt{\frac{((l + 1)^2 - m^2)((l + 2)^2 - m^2)}{(2l + 1)(2l + 3)^2(2l + 5)}}$$

$$\mathfrak{g}_{lk,\bar{1}\bar{2}} = -\sqrt{(k - 1)k(k - 1 + \alpha'_s)(k + \alpha)}$$

$$\mathfrak{g}_{lk,\bar{1}\bar{1}} = 2\sqrt{k(k + \alpha)}(\alpha - 2 - 3\delta_{W,2})$$

$$\mathfrak{g}_{lk,\bar{1}0} = 2k^2 - 4(\alpha - 1 + \delta_{W,2})(\alpha - 3 + 2\delta_{W,2}) + (\alpha + 1)(2k + 3\alpha - 8 + 6\delta_{W,2})$$

$$\mathfrak{g}_{lk,\bar{1}1} = -2\sqrt{(k + 1)(k + 1 + \alpha)}(\alpha - 5 + 3\delta_{W,2})$$

$$\mathfrak{g}_{lk,\bar{1}2} = -\sqrt{(k + 1)(k + 2)(k + 1 + \alpha)(k + 2 + \alpha)}$$

$$\mathfrak{g}_{lk,0\bar{2}} = -\sqrt{(k - 1)k(k + \alpha)}$$

$$\mathfrak{g}_{lk,0\bar{1}} = -\sqrt{k}(k + 3 - 2\delta_{W,2})$$

$$\mathfrak{g}_{lk,00} = \sqrt{k + \alpha + 1}(k + \alpha - 2 + 2\delta_{W,2})$$

$$\mathfrak{g}_{lk,01} = \sqrt{(k + 1)(k + 1 + \alpha)(k + 2 + \alpha)}$$

$$\mathfrak{g}_{lk,1\bar{2}} = -\sqrt{(k - 1)k(k + \alpha)}$$

$$\mathfrak{g}_{lk,1\bar{1}} = -\sqrt{k}(k + 2\alpha + 5 - 2\delta_{W,2})$$

$$\mathfrak{g}_{lk,10} = \sqrt{k + \alpha + 1}(k - \alpha - 4 + 2\delta_{W,2})$$

$$\mathfrak{g}_{lk,11} = \sqrt{(k + 1)(k + 1 + \alpha)(k + 2 + \alpha)}$$

$$\mathfrak{h}_{lk,vs}^{(\kappa)} = (1 - \kappa) \delta_{s0} \delta_{v0} + \kappa \mathfrak{g}_{lk,sv}$$

THE SYMMETRY

Symmetry considerations

States with m even are indexed by **inversion** (ω) and **acyclic** (σ) symmetry.

$$\Psi_{\lambda\eta;m_e l k, \sigma\omega}^{(Wq)} = \Phi_{\lambda\eta;m_e l k, (\sigma i^m)(\omega \sigma i^m (-1)^l)}^{(Wq)}$$

$$\sigma = \pm 1$$

$$\omega = \pm 1$$

States with m odd are indexed by **acyclic** (σ) symmetry and a **free index** (h).
 m odd states only have odd **inversion**.

$$\Psi_{\lambda\eta;m_o l k, \sigma}^{(Wq_1)} = \Phi_{\lambda\eta;m_o l k, h(h(-1)^l)}^{(Wq_2)} - h\sigma i^m \Phi_{\lambda\eta;m_o l k, \bar{h}(h(-1)^l)}^{(Wq_3)}$$

$$\sigma = \pm 1$$

$$h = \pm 1$$

Effect of Td operations, including Bloch functions, m even

States indexed by **inversion** (ω) and **acyclic** (σ) symmetry as either even (+1) or odd (-1)

$$\Psi_{ml;\sigma\omega}^{(q)} = \Phi_{m(\sigma i^m)l}^{(q)} \Theta_{\omega\sigma i^m}^{(q)} (-1)^l$$

$$\Theta^{(q)} = e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{\mathbf{k}_q} \quad \Theta_{\Omega}^{(q)} = \frac{\Theta^{(q)} + \Omega \Theta^{(-q)}}{\sqrt{2}}$$

Cyclic permutations

$$\begin{array}{ccc} (123) & (312) & (231) \\ \Psi_{ml;\sigma\omega}^{(1)} & \Psi_{ml;\sigma\omega}^{(3)} & \Psi_{ml;\sigma\omega}^{(2)} \end{array}$$

Acyclic permutations

$$\begin{array}{ccc} (132) & (321) & (213) \\ \sigma \Psi_{ml;\sigma\omega}^{(1)} & \sigma \Psi_{ml;\sigma\omega}^{(3)} & \sigma \Psi_{ml;\sigma\omega}^{(2)} \end{array}$$

Double inversion

$$\begin{array}{ccc} (\bar{1}\bar{2}\bar{3}) & (\bar{1}\bar{2}\bar{3}) & (\bar{1}\bar{2}\bar{3}) \\ \Psi_{ml;\sigma\omega}^{(1)} & \omega \Psi_{ml;\sigma\omega}^{(1)} & \omega \Psi_{ml;\sigma\omega}^{(1)} \end{array}$$

Time-reversal

$$\omega (-1)^l \Psi_{ml;\sigma\omega}^{(1)}$$

Effect of Td operations, including Bloch functions, m odd

States indexed by **acyclic** (σ) symmetry as either even (+1) or odd (-1). m odd states have odd **inversion** ($\omega = -1$).

$$\Psi_{mlh;\sigma}^{(q_1)} = \Psi_{mlh}^{(-q_2)} - h\sigma i^m \Psi_{mlh}^{(+q_3)}$$

$$\Psi_{mlh}^{(\pm q)} = \Phi_{m(\mp h)l}^{(q)} \Theta_{h(-1)^l}^{(q)}$$

Cyclic permutations

$$\begin{matrix} (123) & (312) & (231) \\ \Psi_{mlh;\sigma}^{(1)} & \Psi_{mlh;\sigma}^{(3)} & \Psi_{mlh;\sigma}^{(2)} \end{matrix}$$

Acyclic permutations

$$\begin{matrix} (132) & (321) & (213) \\ \sigma\Psi_{mlh;\sigma}^{(1)} & \sigma\Psi_{mlh;\sigma}^{(3)} & \sigma\Psi_{mlh;\sigma}^{(2)} \end{matrix}$$

Double inversion

$$\begin{matrix} (1\bar{2}\bar{3}) & (\bar{1}2\bar{3}) & (\bar{1}\bar{2}3) \\ \Psi_{mlh;\sigma}^{(1)} & -\Psi_{mlh;\sigma}^{(1)} & -\Psi_{mlh;\sigma}^{(1)} \end{matrix}$$

Time-reversal

$$(-1)^l \Psi_{mlh;\sigma}^{(1)}$$

Full Accounting of Symmetry, Basis

Multi-valley basis functions are symmetrized according to T_d symmetry of silicon.

$$\Psi_{\lambda\eta;A_1} = \sum_{m_e l k} c_{m_e l k} \sum_{q=x,y,z} \left(\frac{1}{\sqrt{3}} \right) \Phi_{\lambda\eta;m_e l k, (i^{m_e})(i^{m_e}(-1)^l)}^{(q)}$$

$$\Psi_{\lambda\eta;E_n} = \sum_{m_e l k \sigma} c_{m_e l k \sigma} \sum_{q=x,y,z} \left(\frac{\beta_{E_n}^{(\sigma)}}{\sqrt{3}} \right)_q \Phi_{\lambda\eta;m_e l k, (\sigma i^{m_e})(\sigma i^{m_e}(-1)^l)}^{(q)}$$

$$\beta_{E_1}^{(\sigma)} = i^{\frac{1-\sigma}{2}} \{ \tau, \tau^2, 1 \} \quad \beta_{E_2}^{(\sigma)} = \left(\beta_{E_1}^{(\sigma)} \right)^* \quad \tau = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$$

$$\Psi_{\lambda\eta;T_2^{(q)}}^{(e)} = \sum_{m_e l k} c_{m_e l k} \Phi_{\lambda\eta;m_e l k, (i^{m_e})(-i^{m_e}(-1)^l)}^{(q)} \quad \Psi_{\lambda\eta;T_2^{(q_1)}}^{(o)} = \sum_{m_o l k h} c_{m_o l k h} \frac{\Phi_{\lambda\eta;m_o l k, h(h(-1)^l)}^{(q_2)} - h i^{m_o} \Phi_{\lambda\eta;m_o l k, \bar{h}(h(-1)^l)}^{(q_3)}}{\sqrt{2}}$$

Forms for A_2 and T_1 are also known.

Selection rules for all symmetries are analytically known, but have not yet been implemented in code.

POTENTIAL MODEL

Flexible Potential Model

To facilitate deep exploration, **a very general potential model**.

$$U_{\mathbf{r}} = \sum_i \mathcal{U}_i r^{\epsilon_i} \sum_j \mathcal{U}_{ij\{\Omega\}} e^{-\mathcal{A}_{ij\{\Omega\}} r} e^{-\mathcal{B}_{ij\{\Omega\}} r^2}$$

$$\mathcal{F}_{ij\{\Omega\}} = f_{ij,0} + f_{ij,x} \chi_x + f_{ij,y} \chi_y + f_{ij,z} \chi_z$$

$$\chi_x = \frac{x}{r} = \sin(\theta) \cos(\phi) \quad \chi_y = \frac{y}{r} = \sin(\theta) \sin(\phi) \quad \chi_z = \frac{z}{r} = \cos(\theta)$$

- The fraktur constants can be freely set by users of the software.
- Potential form **represents most extant models** of the effective silicon donor potential.
- Examples include the tetrahedral potential used by Gamble [1] and the dynamic dielectric model [2,3].

$$U_{\text{Gamble}} = -\frac{2}{r} + A_0 e^{-\beta_0 r^2} + A_1 \sum_{i=1}^4 e^{-\beta_1 |\mathbf{r} - b \mathbf{t}_i|^2}$$

$$\mathbf{t}_i = \{[111], [\bar{1}1\bar{1}], [1\bar{1}\bar{1}], [\bar{1}\bar{1}1]\}$$

$$U_{\text{Dyn}} = -\frac{2}{r} \left(1 + A e^{-\alpha r} + (\epsilon_0 - A) e^{-\beta r} - e^{-\gamma r} \right)$$

TEST CASE

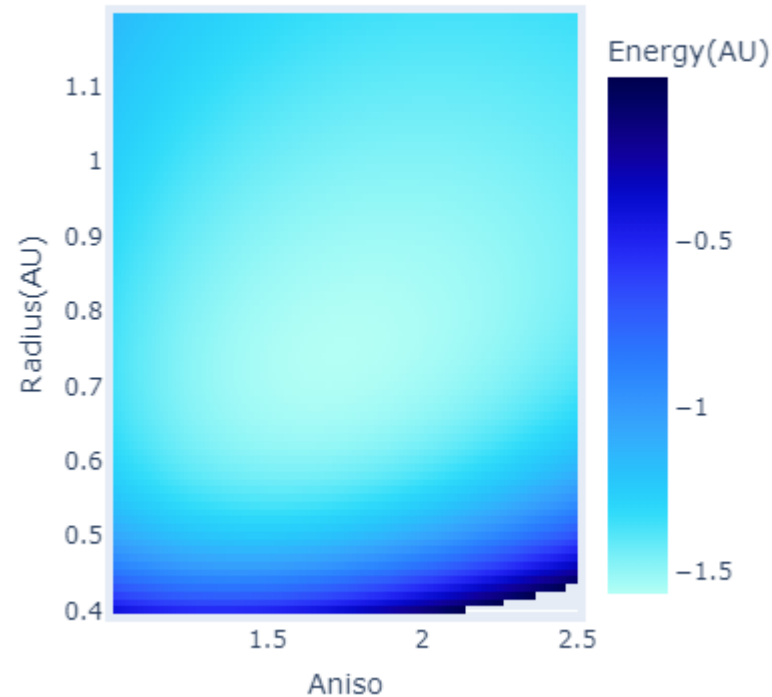
*Single-Valley Wavefunctions
Beyond Kohn-Luttinger*

Kohn-Luttinger (Historic)

$$\epsilon_{AU} = 19.944 \text{ meV}$$

$$\alpha_{AU} = 3.167 \text{ nm}$$

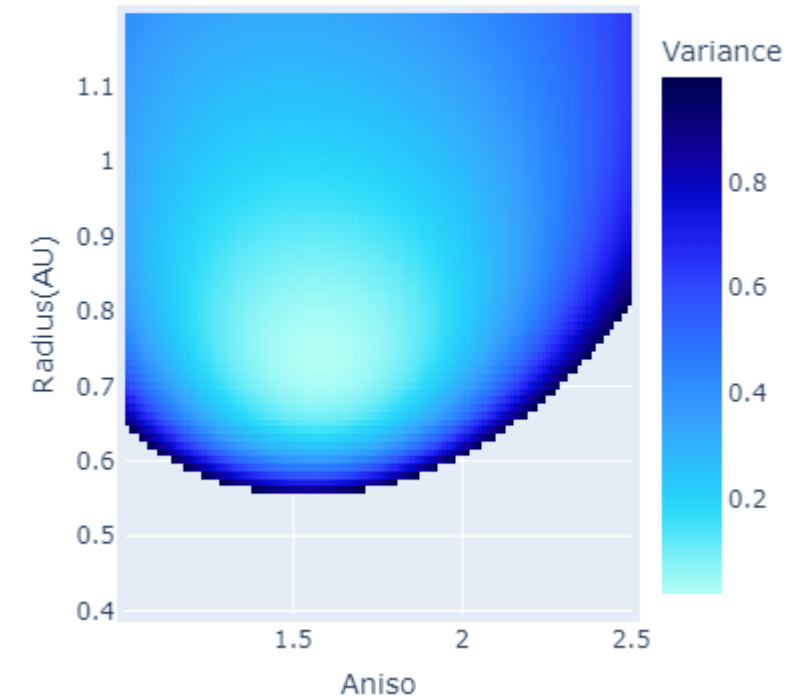
Basis(m,l,k)
0,0,0



$$\lambda \rightarrow 1.745 \quad \eta \rightarrow 0.7522$$

$$E[\lambda, \eta] = -1.565 \quad v[\lambda, \eta] = 0.05048$$

$$E[\lambda, \eta] = -31.22 \text{ meV}$$



$$\lambda \rightarrow 1.585 \quad \eta \rightarrow 0.7421$$

$$E[\lambda, \eta] = -1.558 \quad v[\lambda, \eta] = 0.01877$$

$$E[\lambda, \eta] = -31.08 \text{ meV}$$

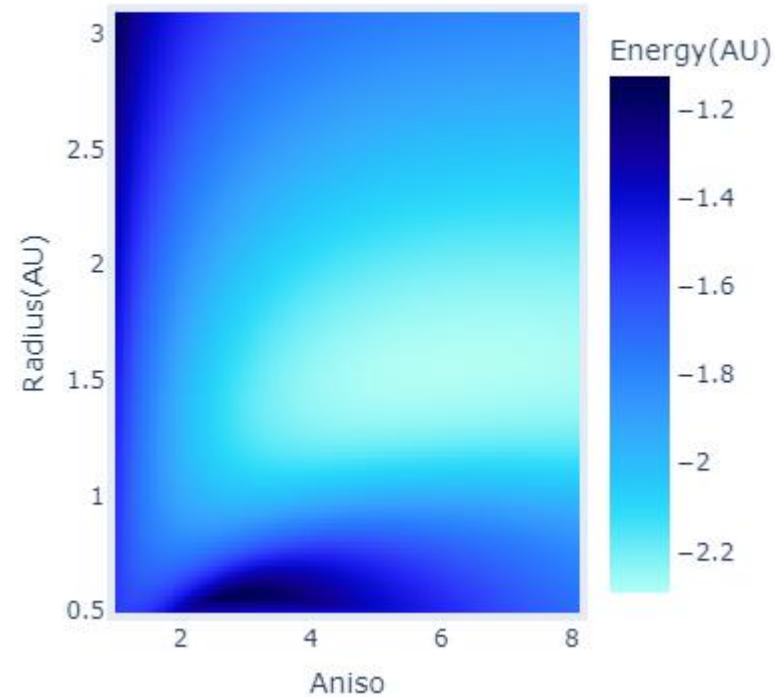
Expanded Single-Valley

$$\epsilon_{AU} = 19.944 \text{ meV}$$

$$a_{AU} = 3.167 \text{ nm}$$

Basis(m,l,k)

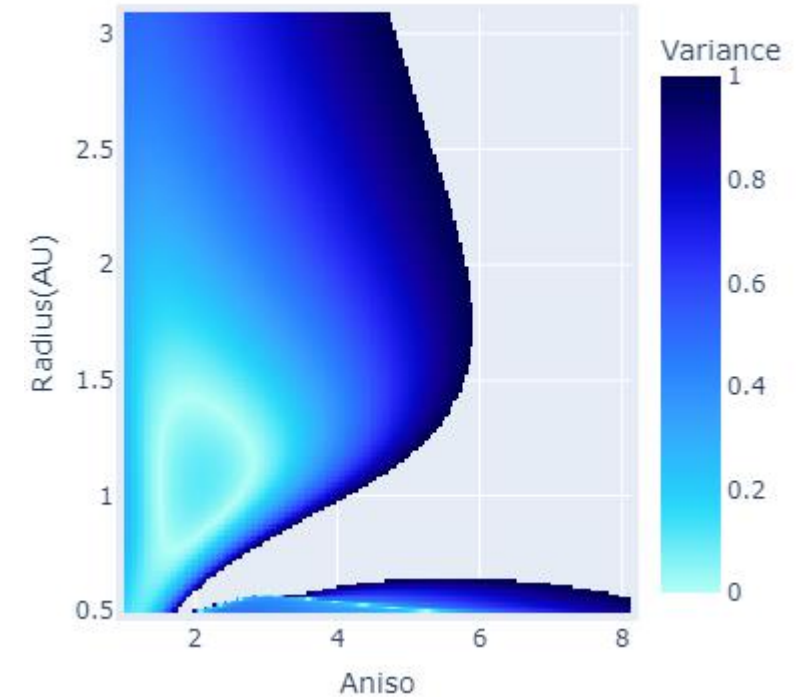
0,0,0
0,0,1
0,0,2
0,0,3
0,2,0
0,2,1
0,2,2
0,2,3
0,4,0
0,4,1
0,4,2
0,4,3
0,6,0
0,6,1
0,6,2
0,6,3



$$\lambda \rightarrow 6.130 \quad \eta \rightarrow 1.558$$

$$E[\lambda, \eta] = -2.294 \quad v[\lambda, \eta] = 1.092$$

$$E[\lambda, \eta] = -45.74 \text{ meV}$$



$$\lambda \rightarrow 1.628 \quad \eta \rightarrow 1.349$$

$$E[\lambda, \eta] = -1.832 \quad v[\lambda, \eta] = 0.00001905$$

$$E[\lambda, \eta] = -36.54 \text{ meV}$$

CONCLUSIONS

Caveats and Challenges -- I

Precalculations

- Complete basis requires computation of alternating sign sums – floating point computation unstable.
- Alternating sign computations performed symbolically with python module Sympy.
- Results stored in SQLite database; retrieved at runtime.
- *Symbolic precalculation takes substantial time, but a one-time investment.*

Database Access

- Size of the precalculation SQLite database on the order of GBs.
- Network distribution difficult. *User assumes computation and storage responsibilities.*
- *Importation into runtime memory presents a recurring bottleneck.*

Caveats and Challenges -- II

Parameter Optimization

- Variational tuning of subbases currently requires hand-tuning.
- *Superbasis* concept introduces large multi-dimensional variational space, requires stochastic search.
- Earlier proof of concept in Fortran implemented an Artificial Bee Colony stochastic optimization algorithm.
- ***Working on integrating extant ABC code within current incarnation of project.***

Numerical Stability and Verisimilitude

- Computations involve numerical matrix and cubature methods.
- Some regions of parameter space seem “sour”/“unstable” leading to unreliable or unbelievable results.
- Multi-valley overlap, kinetic elements necessary for **completeness** and **consistency**, but detracts from **verisimilitude**.
- ***Working on expanding unit testing of code, as well as runtime monitors for numerical stability.***
- ***Working on runtime analysis on breakdown and deviations from the EMA.***

Prospects

- Specific focus of my PhD research on Si:P and the EMA, but *framework is very general*.
- Want to make code and documentation available as an open science, open notes, open source GPL 3 licensed project.

Can be used to explore other Hamiltonians, other symmetries, other systems.

Extensions

- Perturbations of T_d symmetry.
- External fields, electric, magnetic.
- Multi-center effects.
- Multi-electron effects.

Forks

- Different donors in silicon.
- Different bulk materials.
- 2-D systems: quantum dots.

“The Pitch”

Upshot: Want others to benefit from 10,000+ lines of code, thousands of work hours.

- **Opportunity for a collaboration led by PhD students, postdocs, early career scientists.**
- **A chance to develop scientific, technical, logistic, communication, and management skills.**
 - **Python:** Symbolic computations, data analysis, data visualization, SQL integration.
 - **C++:** Code analysis, optimization, documentation. Work with new features of latest C++ standard.
 - **Fortran:** Integration of scientific code from multiple languages.
 - **Mathematics:** Symmetry group theory. Numerical stability. Stochastic optimization. Function representation.

Interested?

- E-mail: lukepend@buffalo.edu, luke.pendo@gmail.com
- Gitlab repo: cantankerous-impurity